MINI PROJECT | EVALUATION 1

Real-Time Network Intrusion Detection System

using Packet Inspection and Web-Based Alerting

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Problem Statement

Small networks have no real-time way to see attacks as they happen.

Where: homes, hostels and college labs

Today: attacks pass unseen in the packets

Gap: tools are manual, complex or unexplainable

Need: light, explainable, alerts within a second

ATTACKS THIS SYSTEM TARGETS

1

Port Scan

Tries every door to find an open one

2

SYN Flood

Floods a server with half-open connections

3

ARP Spoofing

Puts the attacker in the middle of your traffic

Objectives

1 Done

Capture live packets

TCP, UDP and ARP from Wi-Fi

2 Done

Detect port scans

More than 15 ports in 5 seconds

3 Done

Live web dashboard

Counts and red alerts, no refresh

4 Next

Detect SYN floods

Many SYNs, no completed handshakes

5 Next

Detect ARP spoofing

Gateway IP with a new MAC address

6 Next

Test and measure

Detection time and false alarms

Existing Systems and the Gap

Existing tool	What it does well	Gap for small networks
Wireshark	Captures and decodes every packet	Manual inspection, no automatic alerts
Snort, Suricata	Professional rule-based detection	Complex rules, expert setup, log files
Zeek	Deep analysis with scripts	Steep learning curve, no live dashboard
ML-based IDS	High accuracy on test datasets	Black box, needs training data

PROPOSED SYSTEM

Lightweight

About 150 lines, runs on a laptop

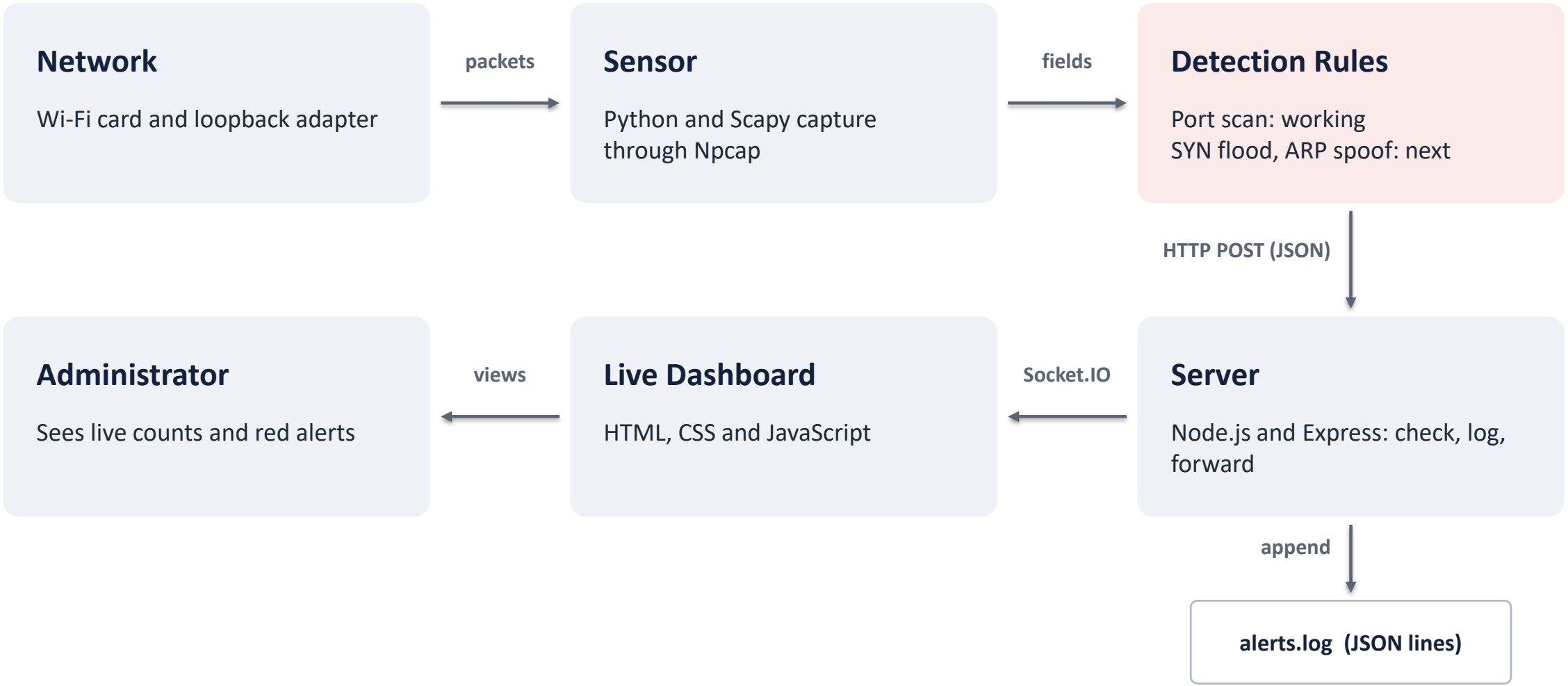
Explainable

Every alert shows its threshold

Live

Alert on the dashboard in under 1 second

System Architecture



Technology Stack

CAPTURE

Npcap 1.89

Reads raw packets on Windows

SENSOR

Python 3.13 and Scapy 2.8

Decodes IP, TCP, UDP and ARP

RULES

Python standard library

deque window and set of ports

SERVER

Node.js 24 and Express 5

Receives alerts by HTTP POST

LIVE PUSH

Socket.IO 4

Alerts reach the browser instantly

DASHBOARD

HTML, CSS, JavaScript

Runs in any browser

Storage: alerts.log, SQLite planned

Testing: scan_test.py, Nmap, Wireshark

Tools: VS Code, Git

Methodology

Development model: Incremental, each round adds a working module

Round 1

Done

Till 10 October

Capture, port scan, server, dashboard

Round 2

Next

Till 30 October

SYN flood and ARP spoofing rules

Round 3

Next

November

Testing, report and final demo

How the system works

1 → 2 → 3 → 4 → 5

Capture

Scapy and Npcap with a packet filter

Count

TCP, UDP and ARP every second

Detect

Rules run on every packet

Deliver

HTTP POST to Express, saved to log

Display

Socket.IO pushes to the dashboard

Design: Data Flow Diagrams

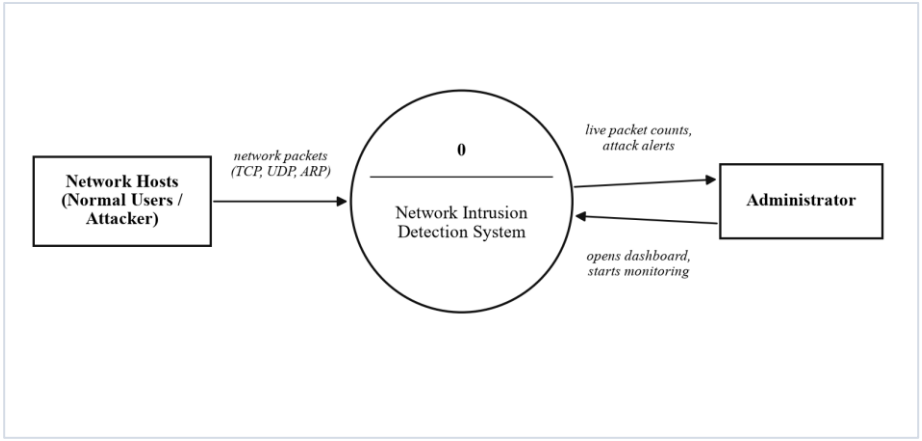


Fig. 3.1: Level 0 DFD, context diagram

Level 0: the whole system as one process

Level 1: five processes and two data stores

Data stores: SYN history and alerts.log

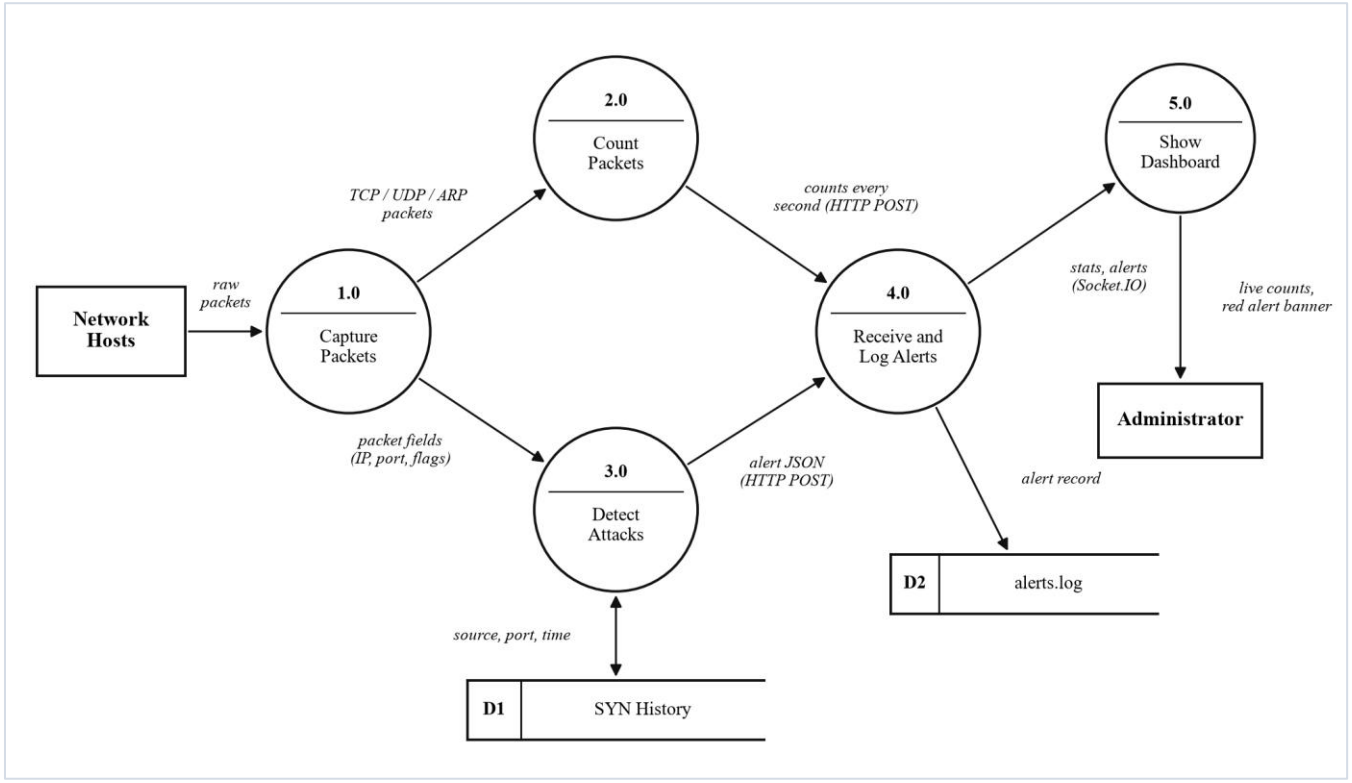


Fig. 3.2: Level 1 DFD

Design: Use Case and Data Storage

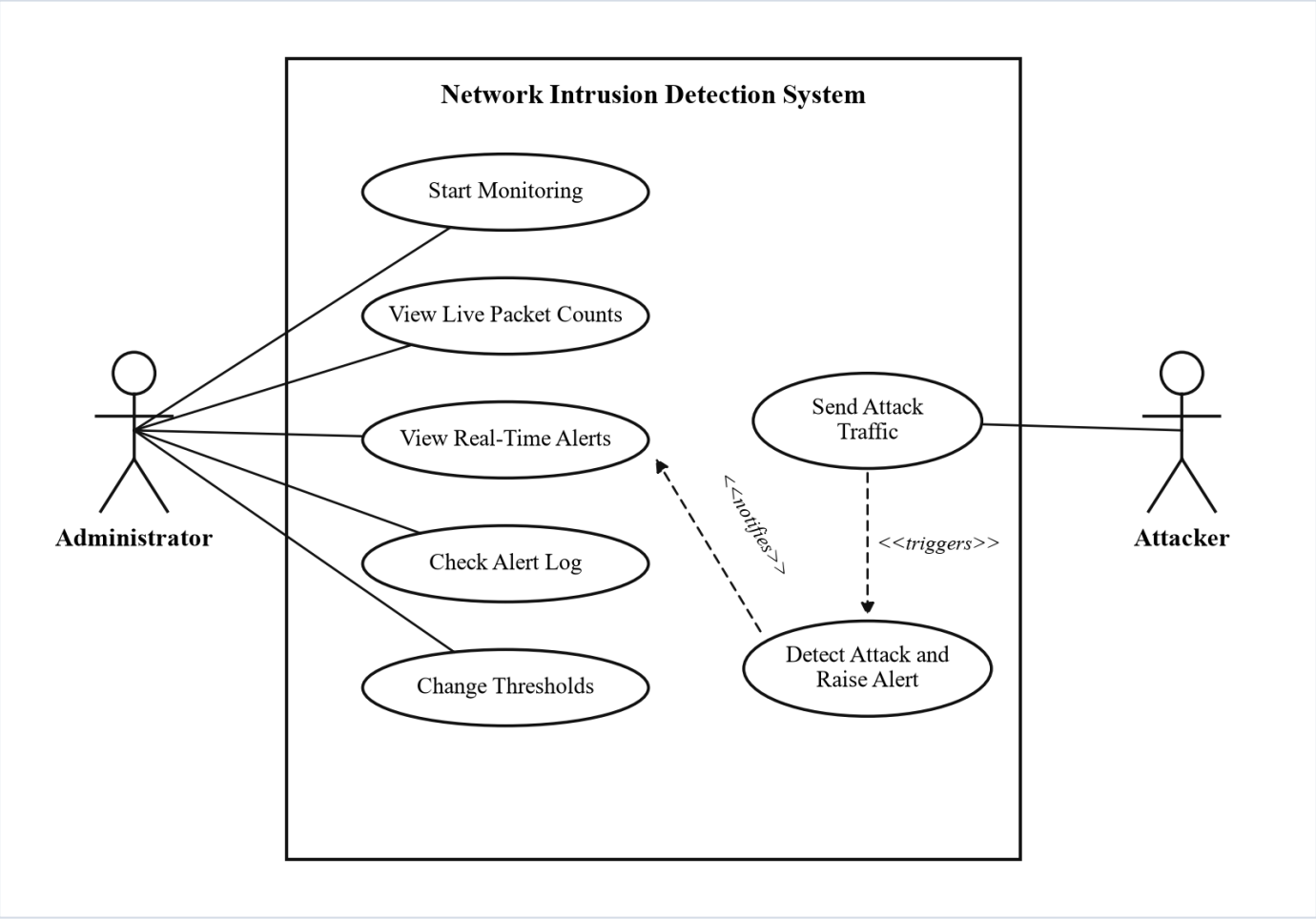


Fig. 3.3: Use case diagram

Administrator: starts monitoring, watches counts and alerts, checks the log

Attacker: sends traffic that triggers detection

System: detects the attack and notifies the admin

DATA STORAGE

SYN history: in memory, last 5 seconds

Alerts: alerts.log, one JSON line each

Planned: SQLite for alert search

Port Scan Detection Logic

Alert when one IP sends SYN packets to more than 15 different ports within 5 seconds

15	5 s	30 s
port limit	sliding window	cool-down per IP

Why SYN only: a scanner knocks on every port with a SYN

Why a sliding window: a scan split across time is never missed

Why a cool-down: one scan gives one alert, not hundreds

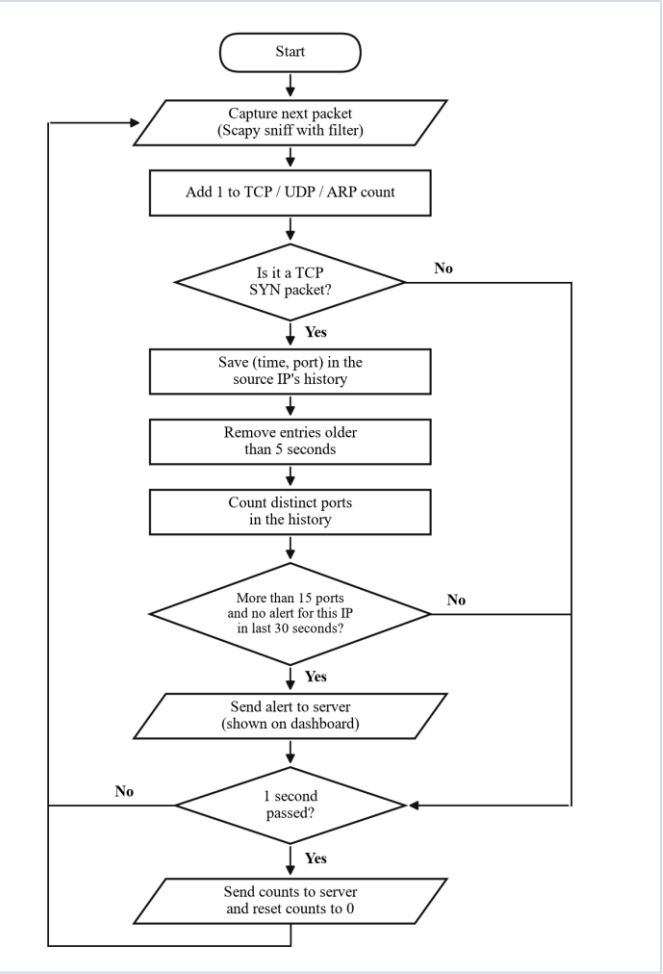
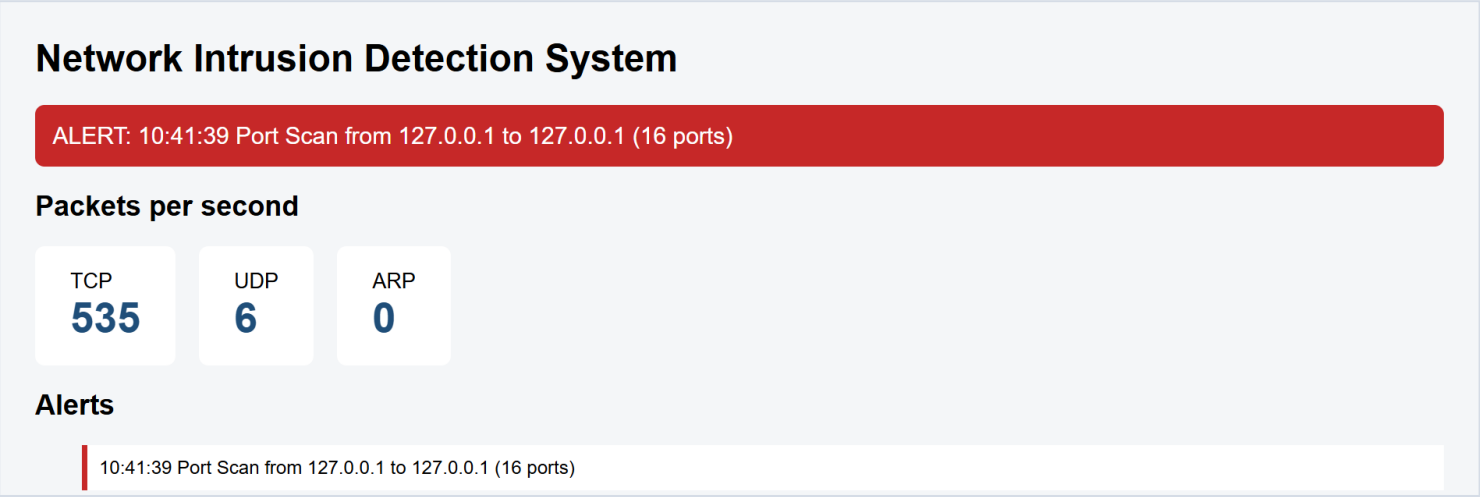


Fig. 3.4: Port scan detection flowchart

Progress and Live Demo



Real dashboard during a test scan, 10 October 2026

50%

of the system is working
Full pipeline with 1 of 3 attack rules

Demo: start server, start sensor, run a test scan, see the red alert

DONE

- Live capture on Wi-Fi and loopback
- Packet counts every second
- Port scan detection
- Alert log and live dashboard

NEXT, BY 30 OCTOBER

- SYN flood rule
- ARP spoofing rule
- Alert history on page reload
- Testing from a second device

Thank You

Questions are welcome

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